

ONC-305-301

Statistical Analysis Plan

Pharma-style simulated version with estimands, TLF traceability and QC plan

Document	Statistical Analysis Plan	Study	ONC-305-301
Version	2.0	Status	Final simulated package
Data cut-off	2026-01-31	Database lock	2026-02-15
Author	Luke Johnston	Classification	Portfolio demonstration
Purpose	Biostatistics/statistical-programming portfolio	Data	100% simulated; no patient data

Important: this is a simulated portfolio package designed to demonstrate biostatistics, statistical programming, reproducibility, and regulatory-document writing. It is not a real clinical trial and is not clinical evidence.

Document administration

Version	Date	Author	Description
1.0	2026-02-01	Luke Johnston	Initial simulated SAP with endpoint and population definitions
2.0	2026-02-15	Luke Johnston	Professional version adding estimand, censoring, sensitivity, TLF traceability and QC sections

Role	Name	Signature	Date
Study Statistician	Luke Johnston	_____	_____
Statistical Programming Lead	Portfolio demonstration	_____	_____
Clinical Lead	Simulated	_____	_____
Data Management Lead	Simulated	_____	_____

In a real sponsor/CRO setting, this section would be finalized before database lock and signed according to the study's document-control process. It is included here to demonstrate awareness of formal SAP governance.

Abbreviations

Term	Meaning	Term	Meaning
ADaM	Analysis Data Model	ADTTE	Analysis dataset for time-to-event endpoints
AE/TEAE	Adverse event / treatment-emergent adverse event	BOR	Best overall response
CNSR	Censoring indicator	CSR	Clinical study report
DCR	Disease control rate	FAS	Full Analysis Set
HR	Hazard ratio	KM	Kaplan-Meier
ORR	Objective response rate	PFS/OS	Progression-free survival / overall survival
SAP	Statistical Analysis Plan	TLF	Tables, listings and figures

1. Study design and statistical objectives

Design feature	SAP specification
Study design	Randomized, double-blind, placebo-controlled Phase III oncology simulation
Subjects	420 randomized; 211 ONC-305 + SOC and 209 Placebo + SOC
Randomization	1:1 allocation with stratification by ECOG performance status and PD-L1 category
Primary objective	Compare PFS between treatment arms in the FAS
Key secondary objectives	Compare OS and ORR; characterize exposure and safety
Data cut-off	2026-01-31
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The SAP is written from the perspective of a statistician preparing a final analysis package. It defines the analysis intent before interpreting treatment results and links each major endpoint to analysis datasets and outputs.

2. Endpoints and endpoint derivations

Endpoint	Dataset / variable	Derivation	Output
PFS	ADTTE; PARAMCD=PFS; AVAL; CNSR	Months from randomization to progression/death/censor date; CNSR=0 event	T14.2.1, F14.2.1
OS	ADTTE; PARAMCD=OS; AVAL; CNSR	Months from randomization to death/last-known-alive date; CNSR=0 death	T14.2.2, F14.2.2
ORR	ADRS; ORRFL	Y if BOR is CR or PR; all FAS subjects in denominator	T14.2.3
DCR	ADRS; DCRFL	Y if BOR is CR, PR or SD; all FAS subjects in denominator	T14.2.3
Exposure	ADSL; TRTDURM; RDI	Duration and relative dose intensity from synthetic exposure records	T14.1.3
Safety	ADAE; ADLB	Subject-level TEAE, grade, seriousness, discontinuation, fatal and lab-shift summaries	T14.3.x, L16.2.x

3. Estimands

The estimand section defines the treatment effect of interest before analysis. This is now a key expectation in modern clinical-trial planning and is especially important for missing data and intercurrent-event handling.

Attribute	Primary PFS estimand	Key secondary OS estimand	Response estimand
Treatment	ONC-305 + SOC vs Placebo + SOC	ONC-305 + SOC vs Placebo + SOC	ONC-305 + SOC vs Placebo + SOC
Population	All randomized subjects in FAS	All randomized subjects in FAS	All randomized subjects in FAS
Variable	Time to progression/death	Time to death	Confirmed or simulated best overall response
Intercurrent events	Treatment-policy for discontinuation/subsequent therapy; death before progression is event	Death regardless of treatment discontinuation	Missing/non-evaluable response remains in denominator
Summary measure	Cox HR; KM median and landmark rates	Cox HR; KM median and landmark rates	Risk difference/ORR with exact CI
Sensitivity	Adjusted and alternative censoring analyses	Adjusted/unadjusted Cox summaries	Supportive response-evaluable analysis

4. Analysis populations

Population	Definition	Treatment grouping	Primary outputs
Randomized Set	All randomized subjects	Randomized arm	Disposition and randomization accountability
Full Analysis Set	All randomized subjects	Randomized arm	PFS, OS, ORR, DCR, subgroup analyses
Safety Set	All subjects receiving at least one dose	Actual treatment	Exposure, AE and lab analyses
Response-Evaluable Supportive Set	Subjects with measurable disease and adequate response information	Randomized arm	Supportive response summaries only

5. General analysis conventions

- Unless otherwise stated, all tests are two-sided and confidence intervals are 95%.
- Percentages are based on the relevant analysis population denominator shown in the table header.
- Time-to-event variables are summarized in months using days / 30.4375.
- A subject is counted once per AE summary category using occurrence flags; maximum severity is used where applicable.
- All results in this portfolio are simulated and should be evaluated as evidence of workflow competence rather than product efficacy or safety.

6. Efficacy analysis methods

Analysis	Primary method	Model/covariates	Interpretation
PFS	Stratified/log-rank comparison and Cox model	Treatment main effect; adjusted model includes ECOG and PD-L1 category	HR < 1 favors ONC-305 + SOC
OS	Log-rank comparison and Cox model	Treatment main effect; adjusted model includes ECOG and PD-L1 category	HR < 1 favors ONC-305 + SOC
ORR/DCR	Exact binomial CI by arm	No formal model in primary portfolio	Higher percentage favors ONC-305 + SOC
Subgroups	Descriptive Cox HR within subgroup	Age, sex, ECOG, stage, PD-L1 and region	Hypothesis-generating only

PFS censoring rules

Clinical scenario	Analysis date	CNSR	Rationale
Documented progression	First progression date	0	Primary PFS event
Death before progression	Death date	0	Death is included in PFS
Alive/no progression at cut-off	Last adequate disease assessment	1	No event observed
Withdrawal/lost to follow-up	Last adequate disease assessment	1	Censored before missing follow-up
No post-baseline assessment	Randomization/baseline date	1	Conservative censoring; sensitivity analysis planned

7. Multiplicity, sensitivity and supplementary analyses

Test sequence	Endpoint	Decision rule in simulated portfolio
1	PFS	Primary endpoint at two-sided alpha 0.05
2	OS	Key secondary endpoint interpreted after PFS
3	ORR	Key secondary endpoint interpreted after OS
Supportive	DCR/subgroups/safety	Descriptive only; no confirmatory alpha allocation

Sensitivity/supplementary analysis	Purpose	Planned implementation
Adjusted Cox model	Assess robustness to baseline prognostic factors	Treatment + ECOG + PD-L1 category
Unstratified Cox model	Assess model dependence	Treatment-only Cox model
Alternative PFS censoring	Assess impact of no post-baseline assessment	Censor at randomization or exclude as supportive only
Subgroup analysis	Evaluate consistency	Forest plot with HRs by age, sex, ECOG, stage, PD-L1 and region
Response-evaluable supportive ORR	Assess impact of NE responses	Exclude NE from denominator in supportive table if implemented

8. Safety analysis methods

- Safety summaries use the Safety Set and actual treatment received.
- TEAEs are events starting or worsening on or after first dose in the synthetic data.
- AE tables count subjects, not repeated events, unless the listing explicitly shows event-level records.
- Laboratory shift tables summarize baseline grade to worst post-baseline grade by parameter.
- Fatal AEs, serious AEs and AEs leading to discontinuation are listed for reviewer traceability.

9. Data standards and derivation traceability

Layer	Artifacts	Purpose
Synthetic source	data/raw/*.csv	Represent SDTM-like source records for subjects, AE, labs, exposure, response and TTE
Analysis datasets	data/adam/*.csv	ADaM-style ADSL, ADTTE, ADAE, ADRS and ADLB
Metadata	data/specs and metadata/*.csv	Variable definitions, define-like XML, result metadata and source-to-analysis traceability
Outputs	outputs/tables, outputs/listings, outputs/figures	TLFs for efficacy, safety and listings
Reports	docs and outputs/reports	SAP, CSR-style report, TLF packet, reviewer guide and validation report

10. Programming and quality control

QC area	Expectation	Portfolio evidence
Reproducibility	All outputs regenerated from one command	programs/python/run_all.py
Subject counts	FAS/Safety counts reconcile across datasets	qc/validation_checks.csv
Endpoint integrity	PFS/OS records and CNSR values checked	ADTTE validation checks
Safety integrity	AE chronology and toxicity grades checked	ADAE validation checks
Metadata	Analysis variables documented	adam_spec.csv and define_like_metadata.xml
Reviewer traceability	Outputs linked to source datasets/programs	analysis_results_metadata.csv

11. Planned TLF package

Result ID	Title	Source	Population	Method
T14.1.1	Demographics and baseline characteristics	ADSL	FAS	Counts, percentages, mean/SD, median/range
T14.1.2	Subject disposition	ADSL	FAS	Subject-level disposition counts
T14.1.3	Extent of exposure	ADSL	Safety	Descriptive exposure summaries
T14.1.4	Major protocol deviations	Mock deviation log + ADSL	FAS	Counts and percentages by deviation category
T14.2.1	Primary analysis of PFS	ADTTE	FAS	Kaplan-Meier, log-rank, Cox model
T14.2.2	Overall survival	ADTTE	FAS	Kaplan-Meier, log-rank, Cox model
T14.2.3	Best overall response	ADRS	FAS	CR/PR/SD/PD/NE counts; exact CI for ORR/DCR
F14.2.1	Kaplan-Meier plot for PFS	ADTTE	FAS	KM curve with censoring
F14.2.3	PFS subgroup forest plot	ADTTE + ADSL	FAS	Cox HR by subgroup
T14.3.1	Safety overview	ADAE + ADSL	Safety	Subject-level TEAE categories
T14.3.2	TEAEs by SOC and PT	ADAE	Safety	Counts once per subject per SOC/PT
T14.3.4	Laboratory shift summary	ADLB	Safety	Baseline-to-worst grade shift counts
L16.2.1	Deaths listing	ADAE + ADSL	Safety	Subject listing

12. Software, limitations and references

- Primary executable implementation: Python with pandas, scipy, statsmodels, matplotlib and reportlab.
- SAS and R files are included as templates to demonstrate mapping to regulated workflows; they are not executed by this environment.
- CSV outputs are used for portability. The optional R script can be extended to export XPT files if local dependencies are available.
- No real patients, sites, product, sponsor or clinical claims are represented.

Reference	How it informed this portfolio
ICH E9(R1)	Estimands, intercurrent events and sensitivity-analysis framing
ICH E3	CSR-style reporting structure and appendices
CDISC ADaM	Analysis dataset traceability and reviewability
FDA Study Data Technical Conformance Guide	Study-data submission packaging concepts and define-like metadata awareness